

CSC360 - Computer Graphics and Digital Image Processing

8/9/26 Reflection

Topic: Reflection, Exceptions, Generics, Collections, GUI, Events Handling, and Master-Details Layouts

Today's lesson gave me some ideas about a number of concepts in connection with Java programming and GUI based programs. First of all, we dealt with reflection in programming languages and in Java in particular. I found out that reflection provides the possibility for a program to reflect on its classes, methods, fields, and objects at run time.

We talked about exceptions in programming. I realized that exceptions are situations that can arise in a program, including incorrect input, division by zero, or accessing non-existing content. Exceptions do not make the entire program crash, and they help programmers handle errors correctly. We also talked about runtime and checked exceptions. Runtime exceptions happen during the process of execution, while checked exceptions need to be either handled or declared.

It was also fascinating to know that exceptions are presented differently in software. Software uses error messages, dialogs, logs, and other user-friendly ways to present exceptions rather than displaying the technical data of an error. It gave me an idea that exception handling is not only related to programming but also to the creation of user-friendly interfaces.

Also, the class touched on generic programming and collections. Generic is reusable and type-safe while Collections include list, set, and map, and are helpful in organizing data. In addition, we covered the view as a read-only data and the distinction between console-based and graphical interface.

Furthermore, we learned that through events handling the GUI applications are able to handle user actions like clicking, selection and slider manipulation among others. An event has a source which produces an event; for example, buttons and

checkboxes. Moreover, we learned that checkboxes can handle multiple selections while radio buttons are helpful in choosing one out of many options.

Moreover, the topic of sliders was covered in class, which are helpful in selecting values in a certain range. For example, sliders can be used for volume, brightness, and zoom among others. Dialog boxes, on the other hand, can be used for message, error, confirmation or for taking input from the user.

Key learning

In general, this class has allowed me to see how the use of exceptions, generics, collections, events, and GUIs all tie into the building of a useful application using Java.